Can you observe any pattern in the above table? There is a pattern in the shift of decimal points of the products in the table. In $2.35 \times 10 = 23.5$ the digits are the same, that is 2, 3, 5. Observe 2.35 and 23.5. To which side has the decimal point been shifted, right or left? The decimal point is shifted to the right by one place. Also note that 10 is one zero followed by 1.

In $2.35 \times 100 = 235.0$, observe 2.35 and 235. To which side and by how many digits has the decimal point been shifted? The decimal point has been shifted to the right by two places. Note that 100 is two zeros followed by 1.

Similarly in $2.35 \times 1000 = 2350.0$, we see that the decimal point has been shifted to the right by three places by adding one more digit 0 to the number 235. Note that 1000 is three zeros followed by 1.

Try these

Find

(i) 9.13×10

(ii) 9.13×100

(iii) 9.13×1000

So we conclude that when a decimal number is multiplied by 10, 100 or 1000, the digits in the product are same as in the decimal number but the decimal point in the product is shifted to the right by as many places as there are zeros followed by 1.

Based on these observations we can now say,

$0.02 \times 10 = 0.2$; $0.02 \times 100 = 2$ and $0.02 \times 1000 = 20$.

Can you find the following?

$2.76 \times 10 = ?$

$2.76 \times 100 = ?$

$2.76 \times 1000 = ?$

Example 1.21 Find the value of the following

(i) 3.26×10 (ii) 3.26×100 (iii) 3.26×1000

(iv) 7.01×10 (v) 7.01×100 (vi) 7.01×1000

Solution

(i) $3.26 \times 10 = 32.6$

(ii) $3.26 \times 100 = 326.0$

(iii) $3.26 \times 1000 = 3260.0$

(iv) $7.01 \times 10 = 70.1$

(v) $7.01 \times 100 = 701.0$

(vi) $7.01 \times 1000 = 7010.0$

Example 1.22 A concessional entrance ticket for students to visit a zoo is ₹12.50. How much has to be paid for 20 tickets?

Solution

Cost of one ticket = ₹12.50

Amount to be paid for 20 students = 12.50×20 = ₹250.00

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